

Math 141, F11

Quiz 3

Section:

Name: Answer

SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (6 points) Evaluate each of the following limits if it exists.

(a) $\lim_{x \rightarrow 2} \frac{1}{\sqrt{x^3 - 2x}}$

$$= \frac{1}{\sqrt{2^3 - 2 \cdot 2}} = \frac{1}{\sqrt{8 - 4}} = \frac{1}{\sqrt{4}} = \frac{1}{2}$$

(b) $\lim_{x \rightarrow 2} \frac{x^2 + x - 6}{x - 2}$

$$= \lim_{x \rightarrow 2} \frac{(x+3)(x-2)}{x-2} = \lim_{x \rightarrow 2} x+3 = 5$$

(c) $\lim_{x \rightarrow +\infty} \frac{2x^2 + 1}{3x^2 - 5}$

$$= \lim_{x \rightarrow +\infty} \frac{\frac{2x^2+1}{x^2}}{\frac{3x^2-5}{x^2}} = \lim_{x \rightarrow +\infty} \frac{2 + \frac{1}{x^2}}{3 - \frac{5}{x^2}} = \frac{2}{3}$$

Problem 2. (4 points) Explain why this function is discontinuous at $x = 0$:

$$f(x) = \begin{cases} e^x & x \geq 0, \\ x & x < 0. \end{cases}$$

Solution: $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} x = 0 \neq$
 $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} e^x = e^0 = 1$
 $\Rightarrow f(x)$ is discontinuous at $x = 0$

